

<https://doi.org/10.1038/s44482-025-00002-6>

Pragmatic lessons for harnessing social media for health research through digital data donation post-API access

Leo Thorbecke^{1,2}✉, Anish K. Agarwal^{2,3,4} & Sharath Chandra Guntuku^{1,2,4}

Social media offers a real-time, ecological, and large-scale data source for health research. Platforms are increasingly restricting access, posing a threat to the future of social media-based health research. We describe two challenges with this data: rapidly changing platforms, which affect use and user behavior, and variations in data availability. To address these challenges, we present four pragmatic strategies for accessing and using social media data, focusing on data donation.

Social media offers a real-time, ecological, and large-scale source for health research, with 68% of American adults actively engaging on various platforms¹. Public posts, reactions, and engagement provide insights into health trends, community perceptions of health topics, and reactions to public health messaging campaigns. Research has investigated the predictive capability of language and images posted to social media across various health conditions^{2–4}, the relationship between health information consumption and health behaviors^{5,6}, and the mental health toll on healthcare workers during public health crises⁷. COVID-19 highlighted the potential for social media data, ranging from Yelp to TikTok, to be used as a large-scale tool for understanding and monitoring community health and health behaviors^{8,9}. In one study, Reddit conversations tracked public perceptions of pandemic severity¹⁰. In another, Twitter data was used to detect emerging COVID symptoms before they were reported by the CDC¹¹.

This stream of real-time, community-level data enables early detection of health threats, responsive public health messaging, and innovative approaches for research. However, as the utility expands, so do the ethical, technical, and infrastructural challenges for responsible and sustainable data use. While social media data has historically been widely available through Application Programming Interfaces (API), social media platforms are moving away from allowing access¹², or charging prohibitively high fees for access¹³. Reduced access threatens the future of digital data and health research, prompting researchers to call for solutions¹⁴. Data donation—where individuals voluntarily download and share their social media archives—offers a viable participant-led solution to sustain access to data in a commercial, unpredictable, and dynamic landscape of social media vendors. Data donation, along with parallel data collection methods, close attention to platform usage and trends, and tailoring protocols to address privacy and ethical factors, may allow researchers to continue health research using social media data. Here, we outline the evolving opportunities and challenges in utilizing social media for health research and

provide pragmatic strategies for ongoing engagement with these valuable data sources.

Challenge 1: shifts in platform use

Social media platforms are constantly evolving, which impacts research feasibility and data continuity. User behavior can be influenced by interface modifications, emerging platforms, or sociopolitical factors.

Case example 1: Twitter acquisition in 2022. Twitter's acquisition and rebranding as X in 2022 altered the landscape of its content and user base. Interface redesigns, shifts in the user base, and new content moderation policies led to shifting engagement, as well as user migration to other platforms^{15,16}. One example of the lasting impact of these changes can be seen in how U.S. audiences perceive X by political party: in 2021, Democrats were more likely to view X as “mostly good for American democracy” than Republicans (47% vs. 17% of Republicans), but in 2025, this perception was reversed, with more Republicans reporting X to be “mostly good for American democracy” (58% vs. 17% of Democrats)¹⁷. This shift represents how platform changes can impact perception, use, and overall engagement.

Researchers using Twitter/X data had to rapidly adjust their methodologies or end studies prematurely as the platform restricted access to data, risking research in areas such as disaster relief and infectious disease surveillance, as well as research on the platform itself¹⁸. In 2023, researchers conducted a study on X post-API, demonstrating rapid changes in popular news accounts influencing discourse around geopolitical events¹⁹; noting the difficulty in conducting this research without API, the authors highlight the risks of cutting off access for research amid significant changes to the platform²⁰. Reduced or restricted access to platforms prevents the study of urgent health-related topics, including false information and public health surveillance.

¹Department of Computer and Information Science, University of Pennsylvania, Philadelphia, PA, USA. ²Center for Insights to Outcomes, University of Pennsylvania, Philadelphia, PA, USA. ³Department of Emergency Medicine, University of Pennsylvania, Philadelphia, PA, USA. ⁴Leonard Davis Institute of Health Economics, University of Pennsylvania, Philadelphia, PA, USA. ✉e-mail: lsthorbecke@gmail.com

Challenge 2: variability in data availability

Social media data availability varies, with some platforms restricting access entirely. Twitter/X, Facebook, and Instagram have curtailed API access, while other platforms that provide access, such as TikTok, have imposed stringent restrictions, even when user consent is provided. These shifts disrupt ongoing studies and prevent future work, requiring new data collection strategies that may increase participant burden.

Case example 2: API data collection on Twitter/X. Historically, Twitter's Academic Research API provided access to vast datasets, allowing researchers to analyze topics ranging from maternal mortality²¹ to opioid use²². For example, in 2020, researchers partnered with Washington state to monitor resident mental health using Twitter data; this real-time data allowed the state Department of Health to identify mental health strain hotspots^{23,24}. Per the Twitter Academic Research API agreement, researchers could retrieve a random stream of tweets daily or access the most recent 3,200 tweets per user timeline²⁵.

In early 2023, Twitter/X transitioned to a paid API model with a tiered pricing structure¹³. The "Basic" tier is priced at \$200 per month, offering limited access to data. The "Pro" tier, at \$5,000 per month, provides more extensive access, including features such as a filtered stream API and full-archive search, but may still be insufficient for research needs. Due to the scope and scale of health research, which often requires large, representative samples over extended time periods, researchers typically need access to high-volume, historical, and filtered data streams. The "Enterprise" tier provides the extensive features, such as full-archive search, enhanced filtering, and high-rate limits, that make longitudinal, large-scale public health analysis feasible. However, the Enterprise tier is reported to start at \$42,000 per month, an unrealistic cost for most academic research teams²⁵. Therefore, health studies relying on Twitter data were discontinued or drastically scaled back due to financial constraints. This example highlights the risks of over-reliance on proprietary data sources.

Case example 3: API data collection on Facebook/Meta. Posts made by individuals on their Facebook timeline have the signal to forecast healthcare encounters²⁶, predict depression diagnoses²⁷, and detect relapse in psychotic disorders²⁸; public Facebook group data has been used for public health research broadly^{29–31}. Previously, researchers could use or create an approved research application to collect Facebook timeline posts, allowing users to seamlessly consent to data sharing and provide exact data elements via a single login.

However, between 2018 and 2021, Meta restricted and revoked data access through applications, even for research studies with explicit participant consent³². Applications are only permitted to collect user data for "digital scrapbooks" that compile the user's posts for personal use³³. This policy shift eliminated a key method for real-time health research, as previously, researchers could collect a wide variety of data, including user posts, likes, photos, and more. This forced researchers to shift to alternative methods, including browser plugins, web scraping, and manual data downloads, introducing logistical and participant retention challenges^{34,35}.

Lessons for health researchers

Prioritize data donation where possible. Data donation provides a practical and participant-centered alternative that can sustain research access to longitudinal social media data when platforms restrict automated data collection. This approach helps support health research projects that rely on digital traces.

For example, Google allows users to manually download their data, which can then be uploaded to a secure research repository. With Google Takeout, users can request a file containing data such as Google search history, YouTube search history, and much more. This file is available for download in the user's Google account; once downloaded, it can be uploaded to a secure portal created by researchers. Facebook, Instagram, and other platforms allow users to complete a similar data download process.

In one case, researchers collected Google search queries through Google Takeout and combined this data with electronic health record data to study searches prior to emergency department (ED) visits; results indicated areas for intervention, based on search content and volume, before patients visit the ED³⁶. Importantly, this data remains available despite platform changes, including policies that may restrict access through API: for example, researchers have used data donation to study video consumption and engagement on TikTok³⁷. Donated data may also serve as an avenue for studying platform features, which impact frequency and purpose of use; for example, donated TikTok data has been used to study how its recommendation algorithm impacts user behavior; this algorithm plays a large role in how TikTok's many users consume information³⁷.

However, manual download processes introduce the need for additional staff time to help with participant data donation; researchers should account for multiple potential failure points³⁸, including:

- Complex navigation: users must log in, locate settings, select the data elements, and initiate downloads.
- Timing: Google Takeout may take minutes to hours to generate the file, depending on the file size requested, while Facebook may take multiple days. Delays increase the risk of participant attrition.

Informed consent for data donation is essential and requires intentionality. Studies indicate that users are not fully aware of all data being collected by digital platforms³⁹, and even when downloading files to donate, the formats are not accessible, and participants may not understand how to open the file and review it before donating^{40,41}. In one study, researchers designed an interactive interface allowing users to explore speech data from Google Assistant prior to donation and evaluated data literacy⁴². Results affirmed the importance of supporting participants: researchers should assist participants in exploring their data so they may understand what is collected and how it may be used or interpreted as participants largely were unaware of data being collected and how it may be used to form comprehensive insights into their lives and routines⁴². Data donation offers a natural setting for informing participants what data is collected, how to access and delete their data, and what their data may reveal about them. Researchers should consider utilizing the data donation process to expand their consent and sharing procedures, thereby helping participants understand their data and build trust with them. This may also reduce attrition, a serious challenge with data donation studies⁴³.

Data donation may introduce generalizability concerns that affect study outcomes and should be addressed. For example, individuals who agree to share their data may differ systematically from those who decline—in privacy concerns, technology literacy, health status, or social media use patterns—potentially limiting the generalizability of findings⁴⁴. Additionally, observer effects can influence study outcomes: when users are aware their social media data is being collected, they may alter their posting behavior, self-censoring or reducing engagement, potentially skewing study findings⁴⁵. Corrections may need to be made to account for these and other measurement and representation challenges⁴⁶.

Clearly define data needs and be prepared to pivot through parallel data collection methods. Given the constantly evolving nature of social media platforms, researchers must anticipate disruptions and design adaptive data collection strategies. A key strategy is to maintain parallel methods for obtaining data—such as combining API access (when available) with data donation to ensure continuity. To mitigate disruptions, researchers should:

- Specify essential data elements: define which metrics (e.g., timeline posts, engagement patterns) are critical for study objectives.
- Confirm data redundancy: ensure these elements are accessible through multiple collection methods (e.g., manual downloads as a backup to API access) and platforms.
- Standardize data formats to address discrepancies between API and manual data, ensuring consistency across datasets.

Researchers must be aware of the variability in data that depends on the collection method. API and manual download methods may differ in available metadata, such as unique user identifiers: for instance, Facebook's API assigns alphanumeric IDs, while manual downloads provide usernames. Discrepancies like these can be difficult to detect and make data management tasks, such as deduplication, more challenging. Researchers must account for these discrepancies when designing their data management systems. Understanding these differences early could help refine the collection and analysis approach.

Different platforms vary in data type, structure, and user behavior, necessitating platform-specific approaches. For example, given TikTok and Instagram's emphasis on visual content, researchers must incorporate computer vision methods, such as image classification and visual sentiment analysis. In contrast, Reddit supports long-form, quasi-anonymous, threaded discussions organized by topic-specific communities ("subreddits"). This structure allows for discourse analysis, but requires methods that account for nested conversational hierarchies and subreddit-specific norms. Researchers must therefore adjust their data preprocessing, feature engineering, and analytic pipelines based on platform characteristics. Generalizability also varies: Reddit users tend to skew younger and more male, while Facebook offers a broader demographic reach. A multi-platform approach improves coverage but increases complexity, requiring harmonization across diverse data structures and user behaviors.

One limitation to pursuing data redundancy may be the lack of research expertise and experience with computational methods in health research; health research teams may not have sufficient experience in utilizing API collection methods or various analytical methods. To address this, researchers may consider interdisciplinary collaboration between computer and information science researchers and health researchers.

Track platform changes and contextualize data trends. Platform changes and updates, including algorithm shifts, user base migration, and content moderation policies, can significantly affect data and study findings³⁸. Researchers should maintain a timeline of major platform changes to align with the data collection phase and adjust analyses to account for shifts in engagement or content visibility (e.g., changes in misinformation policies affecting health discourse).

For example, during the COVID-19 pandemic, Twitter's evolving misinformation policies influenced the volume of vaccine-related discourse, impacting public sentiment analyses. In late 2022, Twitter eliminated its COVID-19 misinformation policy, which had previously flagged or removed posts that contained misleading information about the pandemic⁴⁷. Following this change, researchers observed a surge in anti-vaccine content and misinformation, complicating efforts to distinguish genuine public concern from coordinated disinformation campaigns⁴⁸. Studies relying on data before the policy change might underrepresent the prevalence of misinformation, while post-change analyses risk conflating organic sentiment with policy-induced exposure. Without tracking these policy changes, researchers risk misattributing shifts in data to user-driven behavior rather than platform governance or other external factors.

Address ethical and privacy considerations. As social media platforms impose new restrictions, researchers must navigate evolving ethical considerations around user consent, data privacy, and algorithmic biases. Best practices include:

- Transparent participant consent: in addition to clearly communicating data elements collected, data usage, security protocols, and withdrawal options, support participants in exploring and understanding their data.
- Privacy-preserving methodologies: for participants, allow individuals the option of removing data they do not want to share before transferring it to researchers. For researchers, restrict data collected to the minimum required for the research question, anonymize datasets, and employ secure data storage to safeguard participant information.
- Engagement with policymakers and platforms: advocate for research-

friendly policies that strike a balance between data access and privacy protections.

Researchers should consider additional privacy measures depending on data type (e.g., private messages or images) or platform type (e.g., WhatsApp or Tumblr), where expectations of privacy may be greater than public or semi-public data from sources such as Reddit or Facebook timelines.

Additionally, researchers must also work within local and international data protection laws, presenting opportunities and challenges. For instance, the GDPR in the European Union imposes strict requirements on companies around data availability, making data donation possible; it also requires researchers to comply with rules around data minimization, user consent, and the handling of personal information. These frameworks can shape the design of data donation studies, such as determining what constitutes valid consent, how long data may be stored, and whether it can be transferred internationally. Social media research teams must assess ethical and legal obligations early in the study design phase and ensure institutional review board (IRB) approval aligns with applicable regulations.

While the lessons presented here are grounded in a U.S.-based context, many of them apply broadly across global settings. However, in lower-resource environments, researchers may face limited computing infrastructure, data storage constraints, or lower internet reliability, making the use of real-time APIs or participant-led data donation more challenging. Additionally, social media platform penetration and usage patterns vary across countries, influencing the types of platforms studied and the generalizability of the findings. Researchers should assess the relevance of platforms to their target populations and, where appropriate, triangulate social media insights with data from SMS, WhatsApp, or radio to increase representativeness in lower-resource settings^{49,50}.

Conclusion

Social media remains a powerful yet volatile tool for health research. While its real-time, large-scale nature offers unparalleled insights into public health, researchers must be prepared for shifting platform policies, evolving user behavior, and access restrictions. By embracing pragmatic strategies including data donation, cross-platform approaches, and rigorous ethical safeguards, researchers can continue to leverage social media as a transformative resource in public health. While the lessons presented here are brief, future work should continue expanding on these principles to develop guidance for conducting social media research, as some teams are already providing⁴⁶. In the absence of major policy change protecting social media data availability, researchers can sustain social media health research despite evolving constraints by adopting adaptive strategies, such as data donation.

Data availability

No datasets were generated or analysed during the current study.

Received: 1 May 2025; Accepted: 5 September 2025;

Published online: 21 January 2026

References

1. Gottfried, J. Americans' Social Media Use. *Pew Research Center* <https://www.pewresearch.org/internet/2024/01/31/americans-social-media-use/> (2024).
2. Merchant, R. M. et al. Evaluating the predictability of medical conditions from social media posts. *PLOS ONE* **14**, e0215476 (2019).
3. De Choudhury, M., Morris, M. R. & White, R. W. Seeking and sharing health information online: comparing search engines and social media. in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* 1365–1376 (Association for Computing Machinery, New York, NY, USA, 2014). <https://doi.org/10.1145/2556288.2557214>.

4. Guntuku, S. C., Preotiuc-Pietro, D., Eichstaedt, J. C. & Ungar, L. H. What Twitter Profile and Posted Images Reveal about Depression and Anxiety. *Proc. Int. AAAI Conf. Web Soc. Media* **13**, 236–246 (2019).
5. Carrotte, E. R., Vella, A. M. & Lim, M. S. Predictors of “Liking” Three Types of Health and Fitness-Related Content on Social Media: A Cross-Sectional Study. *J. Med. Internet Res.* **17**, e4803 (2015).
6. Al-Dmour, H., Masa'deh, R., Salman, A., Abuhashesh, M. & Al-Dmour, R. Influence of Social Media Platforms on Public Health Protection Against the COVID-19 Pandemic via the Mediating Effects of Public Health Awareness and Behavioral Changes: Integrated Model. *J. Med. Internet Res.* **22**, e19996 (2020).
7. Agarwal, A. K., Mittal, J., Tran, A., Merchant, R. & Guntuku, S. C. Investigating Social Media to Evaluate Emergency Medicine Physicians' Emotional Well-being During COVID-19. *JAMA Netw. Open* **6**, e2312708 (2023).
8. Southwick, L., Francisco, A., Bradley, M., Klinger, E. & Chandra Guntuku, S. Characterizing Responses to COVID-19 Vaccine Promotion on TikTok. *Am. J. Health Promot.* **37**, 638–645 (2023).
9. Sehgal, N. K. R. et al. Disparities by Race and Urbanicity in Online Health Care Facility Reviews. *JAMA Netw. Open* **7**, e2446890 (2024).
10. Stokes, D. C., Andy, A., Guntuku, S. C., Ungar, L. H. & Merchant, R. M. Public Priorities and Concerns Regarding COVID-19 in an Online Discussion Forum: Longitudinal Topic Modeling. *J. Gen. Intern. Med.* **35**, 2244–2247 (2020).
11. Santosh, R., Schwartz, H. A., Eichstaedt, J., Ungar, L. & Guntuku, S. C. Detecting Emerging Symptoms of COVID-19 using Context-based Twitter Embeddings. In *Proceedings of the 1st Workshop on NLP for COVID-19 (Part 2) at EMNLP 2020* (eds Verspoor, K. et al.) (Association for Computational Linguistics, Online, 2020). <https://doi.org/10.18653/v1/2020.nlpcovid19-2.35>.
12. Bruns, A. After the ‘APocalypse’: social media platforms and their fight against critical scholarly research. in *Disinformation and Data Lockdown on Social Platforms* (Routledge, 2021).
13. About the X API. <https://docs.x.com/x-api/getting-started/about-x-api>.
14. Rathje, S. To tackle social-media harms, mandate data access for researchers. *Nature* **633**, 36–36 (2024).
15. X sees largest user exodus since Musk takeover. <https://www.nbcnews.com/tech/tech-news/x-sees-largest-user-exodus-musk-takeover-rcna179793>.
16. Is Elon Musk's X user exodus real? *Newsweek* <https://www.newsweek.com/how-top-twitter-rivals-fared-since-elon-musk-exodus-1984404> (2024).
17. Faverio, M. & Anderson, M. Republicans and Democrats on X differ over the site's politics and their experiences. *Pew Research Center* <https://www.pewresearch.org/short-reads/2025/06/05/republicans-and-democrats-on-x-differ-over-the-sites-politics-and-their-experiences/> (2025).
18. Ledford, H. Researchers scramble as Twitter plans to end free data access. *Nature* **614**, 602–603 (2023).
19. Caulfield, M., Bayar, M. C. & Aske, A. B. The ‘new elites’ of X: Identifying the most influential accounts engaged in Hamas/Israel discourse. *Center for an Informed Public* <https://www.cip.uw.edu/2023/10/20/new-elites-twitter-x-most-influential-accounts-hamas-israel/> (2023).
20. Caulfield, M. The new Twitter is changing rapidly — study it before it's too late. *Nature* **623**, 225–225 (2023).
21. Abebe, R., Giorgi, S., Tedijanto, A., Buffone, A. & Schwartz, H. A. A. Quantifying Community Characteristics of Maternal Mortality Using Social Media. in *Proceedings of The Web Conference 2020* 2976–2983 (Association for Computing Machinery, New York, NY, USA, 2020). <https://doi.org/10.1145/3366423.3380066>.
22. Sarker, A., Gonzalez-Hernandez, G., Ruan, Y. & Perrone, J. Machine Learning and Natural Language Processing for Geolocation-Centric Monitoring and Characterization of Opioid-Related Social Media Chatter. *JAMA Netw. Open* **2**, e1914672 (2019).
23. Guntuku, S. C. et al. Tracking Mental Health and Symptom Mentions on Twitter During COVID-19. *J. Gen. Intern. Med.* **35**, 2798–2800 (2020).
24. Washington State Department of Health. Behavioral Health Resources and Recommendations. <https://doh.wa.gov/emergencies/covid-19/health-care-providers/behavioral-health-resources>.
25. Stokel-Walker, C. Twitter's \$42,000-per-Month API Prices Out Nearly Everyone. *Wired* (2023).
26. Guntuku, S. C. et al. Variability in Language used on Social Media prior to Hospital Visits. *Sci. Rep.* **10**, 4346 (2020).
27. Eichstaedt, J. C. et al. Facebook language predicts depression in medical records. *Proc. Natl. Acad. Sci. USA*. **115**, 11203–11208 (2018).
28. Birnbaum, M. L. et al. Detecting relapse in youth with psychotic disorders utilizing patient-generated and patient-contributed digital data from Facebook. *NPJ Schizophr* **5**, 17 (2019).
29. White, B. K., Burns, S. K., Carson, J. & Scott, J. A. Mapping breastfeeding and COVID-19 related content and engagement on Facebook: Results from an online social listening study. *Health Promot. J. Aust. Off. J. Aust. Assoc. Health Promot. Prof.* **35**, 242–250 (2024).
30. Xue, H., Zhang, J., Sagae, K., Nishimine, B. & Fukuoka, Y. Analyzing Public Conversations About Heart Disease and Heart Health on Facebook From 2016 to 2021: Retrospective Observational Study Applying Latent Dirichlet Allocation Topic Modeling. *JMIR Cardio* **6**, e40764 (2022).
31. Broniatowski, D. A., Simons, J. R., Gu, J., Jamison, A. M. & Abrams, L. C. The efficacy of Facebook's vaccine misinformation policies and architecture during the COVID-19 pandemic. *Sci. Adv.* **9**, eadh2132 (2023).
32. Thaker, K. Introducing Facebook Graph API v11.0 and Marketing API v11.0. *Meta for Developers* <https://developers.facebook.com/blog/post/2021/06/08/introducing-facebook-graph-api-v11-marketing-api-v11/> (2021).
33. Meta. Permissions Reference for Meta Technologies APIs. *Meta for Developers* <https://developers.facebook.com/docs/permissions/>.
34. Breuer, J., Kmetty, Z., Haim, M. & Stier, S. User-centric approaches for collecting Facebook data in the ‘post-API age’: experiences from two studies and recommendations for future research. *Inf. Commun. Soc.* **26**, 2649–2668 (2023).
35. Perriam, J., Birkbak, A. & Freeman, A. Digital methods in a post-API environment. *Int. J. Soc. Res. Methodol.* **23**, 277–290 (2020).
36. Asch, J. M. et al. Google search histories of patients presenting to an emergency department: an observational study. *BMJ Open* **9**, e024791 (2019).
37. Zannettou, S. et al. Analyzing User Engagement with TikTok's Short Format Video Recommendations using Data Donations. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* 1–16 (Association for Computing Machinery, New York, NY, USA, 2024). <https://doi.org/10.1145/3613904.3642433>.
38. Van Driel, I. I. et al. Promises and Pitfalls of Social Media Data Donations. *Commun. Methods Meas.* **16**, 266–282 (2022).
39. Pins, D., Jakobi, T., Boden, A., Alizadeh, F. & Wulf, V. Alexa, We Need to Talk: A Data Literacy Approach on Voice Assistants. in *Designing Interactive Systems Conference 2021* 495–507 (ACM, Virtual Event USA, 2021). <https://doi.org/10.1145/3461778.3462001>.
40. Symoudis, E. et al. Unlocking personal data from online services: user studies on data export experiences and data transfer scenarios. *Human-Computer Interact.* 1–25 <https://doi.org/10.1080/07370024.2024.2325347>.
41. Bowyer, A. et al. Human-GDPR Interaction: Practical Experiences of Accessing Personal Data. in *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* 1–19 (Association for Computing Machinery, New York, NY, USA, 2022). <https://doi.org/10.1145/3491102.3501947>.

42. Gomez Ortega, A., Bourgeois, J., Hutiri, W. T. & Kortuem, G. Beyond data transactions: a framework for meaningfully informed data donation. *AI Soc* **40**, 1–18 (2025).
43. Ohme, J. & Araujo, T. Digital data donations: A quest for best practices. *Patterns* **3**, (2022).
44. Olteanu, A., Castillo, C., Diaz, F. & Kiciman, E. Social Data: Biases, Methodological Pitfalls, and Ethical Boundaries. *Front. Big Data* **2**, 13 (2019).
45. Saha, K., Gupta, P., Mark, G., Kiciman, E. & De Choudhury, M. Observer Effect in Social Media Use. in *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* 1–20 (Association for Computing Machinery, New York, NY, USA, 2024). <https://doi.org/10.1145/3613904.3642078>.
46. Boeschoten, L., Araujo, T., Ausloos, J., Möller, J. & Oberski, D. A framework for privacy preserving digital trace data collection through data donation. *Comput. Commun. Res.* **4**, 388–423 (2022).
47. The Associated Press. Twitter will no longer enforce its COVID misinformation policy. NPR (2022).
48. Lorenz, T. Twitter ends its ban on covid misinformation. *The Washington Post* (2022).
49. Iacoella, F., Gassmann, F. & Tirivayi, N. Which communication technology is effective for promoting reproductive health? Television, radio, and mobile phones in sub-Saharan Africa. *PLOS ONE* **17**, e0272501 (2022).
50. Garimella, K. & Chauchard, S. Whatsapp Explorer: A data donation tool to facilitate research on WhatsApp. *Mob. Media Commun.* 20501579251326809 <https://doi.org/10.1177/20501579251326809> (2025).

Acknowledgements

This research was funded by NIMHD (R01MD018340) and Penn Medicine Communication Research Institute. The funders played no role in study design, data collection, analysis, and interpretation of data, or the writing of this manuscript.

Author contributions

All authors wrote this article and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Leo Thorbecke.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026